

Chapter four: CS S

What is

CSS?

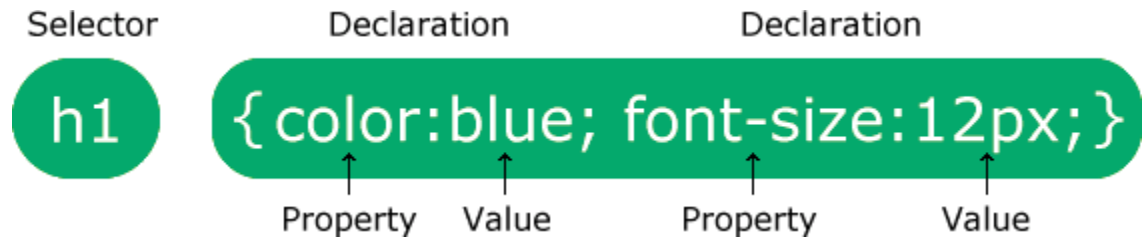
CSS stands for Cascading Style Sheets.

- Designed to specify the overall appearance of web pages as well as the appearance and structure of the text and elements such as images and buttons on web pages and their layout.
- Not just a language all its own, CSS is a part of HTML.
- The first version of HTML to include CSS was HTML 4.0
- CSS was added to HTML to solve a particular problem - the problem of the content of HTML documents not being separated from the layout of the documents.
- The World Wide Web Consortium (W3C) - the organization responsible for developing and maintaining standards on the world wide web created CSS to solve the problem of content not being separated from layout.

What can be done with CSS?

- Specify a background color
- Specify a common style for one tag or a set of tags
- Specify the distance between elements
- Specify the appearance of links
- Specify multiple styles for various webpages

CSS syntax



- There are **three** parts in the CSS syntax:

selector {property: value}

- selector** - The selector is the HTML tag that a style will be applied to. **For example**, the `<p>` tag for paragraphs or the `<h2>` heading tag.
- property** - The property is an characteristic of the HTML tag that will be affected by the stylesheet definition.
For example, the background color of a webpage or the color of links.
- value** - The value that the property of a selector will have. For example, the value of the background color of a webpage can be green or the value of the color of links can be gray.

- The **declaration block** contains one or more declarations separated by semicolons.

Each declaration includes a CSS property name and a value, separated by a colon.

- Multiple CSS declarations are separated with semicolons, and declaration blocks are surrounded by curly braces.
- The property and value of a selector are separated by a colon, and surrounded by curly braces.
- **If a value is more than one word, put quotes around it.**

Example of value being more than one word:

```
h2 {font-family: "Trebuchet MS"}
```

- **If you specify more than one property, each property should be separated with a semicolon.**

Example of multiple properties: `body {background-color: gray; color: yellow;}` ⁵

Selector

Single element selector

Example

```
p{ color: blue;  
    font-family: courier; }
```

Group selector: You can specify that a group of tags will have the same styles by grouping them together.

Example:

```
p, h1, h2, h3, h4 { color: blue;  
                   font-family: courier; }
```

In this example, all text declared with the tags `<p>`, `<h1>`, `<h2>`, `<h3>`, and `<h4>` will be blue in color and Courier in font

Class selector

- Classes allow to define different styles for the same HTML tag / element

For example, say you would like to have two different paragraph types in your web page, one left-aligned and one right-aligned

Selector

type

- **Class selector**

- Classes are declared in CSS by placing a dot in between the element and the class name

For Example:

```
p.Left{  
    color:purple ;  
    Text-align:left;  
}
```

```
p.right{  
    Color:green;  
    Text-align:right;  
}
```

```
<p class="left"> This is some left-aligned text. </p>  
class="right"> This is some right-aligned text. </p>
```

Class Inheritance (cont'd)

```
P{  
    color:blue;  
    Text-align:center  
}
```

```
p.Left{  
    color:purple ;  
    Text-align:left;  
}
```

p
.r
i
g
h

Class Inheritance (cont'd)

- The classes inherit all the properties from the non-class declaration unless a specific property is overwritten by the same property that's also defined in the class
- In this example, the color blue is overwritten with purple in class “left” and with green in class “right”
- If neither of the declarations set a property, the browser default is used

Selector

type Element-less class

- An element-less class is a class that doesn't have an element associated with it.
- These classes can be used by any element
- Element-less classes can be declared using the form:

```
.classname{  
    property:value;  
    property:value;  
}
```

For example we can do the previous example like this

```
.Left{  
    color:purple ;  
    Text-align:left;  
}
```

Selector

type

- **The *#id* selector** styles the element with the specified id.
 - *#id* {property : value;}

Selector

- **type** pseudo-classes :are used to add special effects to some selectors
- It can be used to:
 - Style an element when a user moves the mouse over it
 - Style visited and unvisited links differently
 - Style an element when it gets focus
 - Style valid/invalid/required/optional form

```
elements selector:pseudoclass{  
    property:value  
}
```

- Pseudo-classes can also be combined with classes

```
selector.classname:pseudoclass{  
    property:value  
}
```

Example

```
a.highlight:hover  
{ color:  
  #ff0000;  
}
```

Anchor Pseudo-

classes

- Links can be displayed in different ways:

```
/* unvisited link */
```

```
a:link {
```

```
  color:
```

```
  #FF0000; }
```

```
/* visited link */
```

```
a:visited {
```

```
  color:
```

```
  #00FF00; }
```

```
/* mouse over
```

```
link */
```

```
a:hover {
```

```
  color:
```

```
  #FF00FF; }
```

```
/* selected link
```

```
*/
```

```
a:active {
```

:focus pseudo-class

- The :focus pseudo-class applies styles when the element is focused (e.g., clicked on or navigated to using the keyboard).

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head> <style>
```

```
input:focus {
```

```
    border-color: #4CAF50; /* Green border */
```

```
    box-shadow: 0 0 5px rgba(231, 91, 49, 0.5); /* Subtle green glow */
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
    <form>
```

```
        <label for="name">Name:</label>
```

```
        <input type="text" id="name" placeholder="Enter your name">
```

```
    </form>
```

```
</body>
```

:focus pseudo-class

- You can style form elements based on their validity, requirement, or optional state using the following CSS pseudo-classes:
 - :required: Targets form elements with the required attribute.
 - :optional: Targets form elements without the required attribute.
 - :valid: Applies styles to form elements with valid input.
 - :invalid: Applies styles to form elements with invalid input.

The three types of stylesheets

- ❑ There are three types of stylesheets:

1. **Internal/Embedded** - Placed right on the page whose interface it will affect.

- ❑ **Embedded styles** are styles that are embedded in the head of the document.

- ❑ Embedded styles affect only the tags on the page they are embedded in.

```
<style type="text/css">
```

```
p { color: #00f; } </style>
```

2. **External** - Placed in a separate file.

3. **Inline** - Placed inside a tag it will affect.

- ❑ Inline styles are styles that are written directly in the tag on the document.

- ❑ Inline styles affect only the tag they are applied to.

```
<a href="" style="text-decoration: none;">
```

1. Creating an internal stylesheet

- ❑ Use an internal stylesheet when you want **an HTML document to have a unique style.**
- ❑ An internal stylesheet is defined using the **<style> tag and goes in the head section of an HTML document.**
- ❑ The **<style>** tag specifies the content type of a stylesheet with its **type** attribute which should be **set to "text/css".**

Syntax:

```
<style type="text/css"> styles go here
</style>
```

Example:

```
<html>
<head>
<style type="text/css">
p {color: green}
</style>
</head>
<body>
<p> The text in this paragraph will be green. </p> <p> This
paragraph too. </p>
</body> </html>
```

- ❑ The above stylesheet definition specifies that all text declared with the <p> tag should be green.

Output:

The text in this paragraph will be green.

This paragraph too.

Example:

```
<html>
```

```
<head>
```

```
<style type="text/css">
```

```
<!--
```

```
p {color: green;  
}
```

```
-->
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<p> The text in this paragraph will be green. </p>
```

```
</body>
```

```
</html>
```

2. Creating an external stylesheet

- ☐ Use an external stylesheet when you want to apply one style to many pages.
- ☐ If you make one change in an external stylesheet, the change is universal on all the pages where the stylesheet is used.
- ☐ An external stylesheet is **declared in an external file with a .css extension**. It is called by pages whose interface it will affect.
- ☐ External stylesheets are called using the <link> tag which should
be placed in the head section of an HTML document.

This tag takes three attributes.

Attributes of the <link> tag:

- ❑ **rel** - When using an external stylesheet on a webpage, this attribute takes the value "stylesheet"
- ❑ **type** - When using an external stylesheet on a webpage, this attribute takes the value "text/css"
- ❑ **href** - Denotes the name and location of the external stylesheet to be used.

Example:

```
<html>
```

```
<head>
```

```
<link rel="stylesheet" type="text/css" href="style1.css"  
>
```

```
</head>
```

```
<body>
```

```
<p> The text in this paragraph will be blue. </p>
```

```
</body>
```

```
</html>
```

Output: The text in this paragraph will be blue

The code from style1.css:

```
p {color:blue}
```

NOTE: The <style> tag is NOT used in an external stylesheet, and neither are HTML comments.

3. Creating an inline stylesheet

- ❑ Use inline stylesheets **when you want to apply a style to a single occurrence of an element.**
- ❑ Inline stylesheets are **declared within individual tags and affect those tags only.**
- ❑ Inline **stylesheets are declared with the *style* attribute.**

Example:

❑ `<p style="color: gray">This text will be gray.</p>`

❑ In this example, we are using the stylesheet command **color** to denote that the text in a paragraph will be gray.

Output:

This text will be gray.

Multiple stylesheets

- ❑ You can have **multiple stylesheet definitions on one page.**
- ❑ **For example,** an internal stylesheet and an external stylesheet on one page or an inline stylesheet and an internal stylesheet on one page.
- ❑ If a property has been set for the same tag in different stylesheet definitions on the same page, the definition that will be used will be from the most specific stylesheet.
- ❑ That is, the **stylesheet that has the highest priority.**

Stylesheets by priority:

- ❑ **Inline stylesheet** - An inline stylesheet has the highest priority.
 - ❑ It will override styles declared in an internal stylesheet, an external stylesheet, and a web browsers default styles.
- ❑ **Internal stylesheet** - An internal stylesheet **has the second highest priority**. It will override styles declared in an external stylesheet and a web browsers default styles.
- ❑ **External stylesheet** - An external stylesheet has the third highest priority. It will override a web browsers default styles.
- ❑ **Browsers default styles** - A web browsers default styles **have the lowest priority**. A web browsers default styles will be used when there are no styles set for an element in an inline, internal, or external stylesheet

Example:

```
<html>
```

```
<head>
```

```
  <link rel="stylesheet" type="text/css" href="style2.css" />
```

```
<style type="text/css">
```

```
<!--
```

```
p {color: orange
```

```
}
```

```
-->
```

```
</style>
```

```
</head>
```

```
<body>
```

```
  <p style="color: yellow">
```

```
The text in this paragraph will be yellow. </p>
```

```
</body> </html>
```

- ❑ In this example there are several stylesheet definitions:
- ❑ external stylesheet definition taken from the file style2.css
 - Specifies that the background color of the page should be gray.
- ❑ internal stylesheet definition - Specifies that text declared with the <p> tag should be orange.
- ❑ inline stylesheet definition - Specifies that text declared with one particular <p> tag should be yellow.

- ❑ Assume that this page is viewed in a web browser that has the **default background color for webpages set to light yellow.**
- ❑ The external stylesheet definition will override the web browsers default background color and will set the background color of the page to gray.
- ❑ The inline stylesheet definition will override the internal stylesheets definition that specifies that text declared with the `<p>` tag should be orange, **and will set it to yellow.**

CSS property

1. CSS background properties

- ❑ With CSS background properties, you can define how the background of various elements is displayed.
- ❑ **The focuses are:** Setting a background color & Setting a background image

A. Setting a background color

- ❑ You can use CSS to set the background color of various elements including the webpage itself, tables, textboxes and links.
- ❑ The background color of an element is set with the **background-color property**. You can specify the background color of an element with **a color name, an RGB value, or a hexadecimal value.**

Example:

```
<html> <head> <title>Background properties</title>
<style type="text/css">
body {background-color: #f0f8ff;}
h1 {background-color: lightsteelblue;}
p {background-color: rgb(220, 220, 220);}
a {background-color: #e6dcbe;}
</style>
</head>
<body>
<h1>Some text</h1>
<p> A paragraph </p>
<a href=f="http://www.landofcode.com">
Landofcode.com main page
</a>
</body>
</html>
```

Output: each of the following has different background

Some text

A paragraph

[Landofcode.com main page](#)

B. Setting a background image

- ❑ You can use CSS to set a background image for various elements including the webpage itself and tables.
- ❑ The background image of an element is set with the **background-image** property.
- ❑ The value this property takes should be the location of the image specified within a `url()` identifier.

Example:

```
<html> <head> <title>Setting a background image</title>
```

```
<style type="text/css">
```

```
  body {background-image: url(images/ob020.jpg);}
```

```
  table {background-image: url(images/apple.jpg);}
```

```
</style>
```

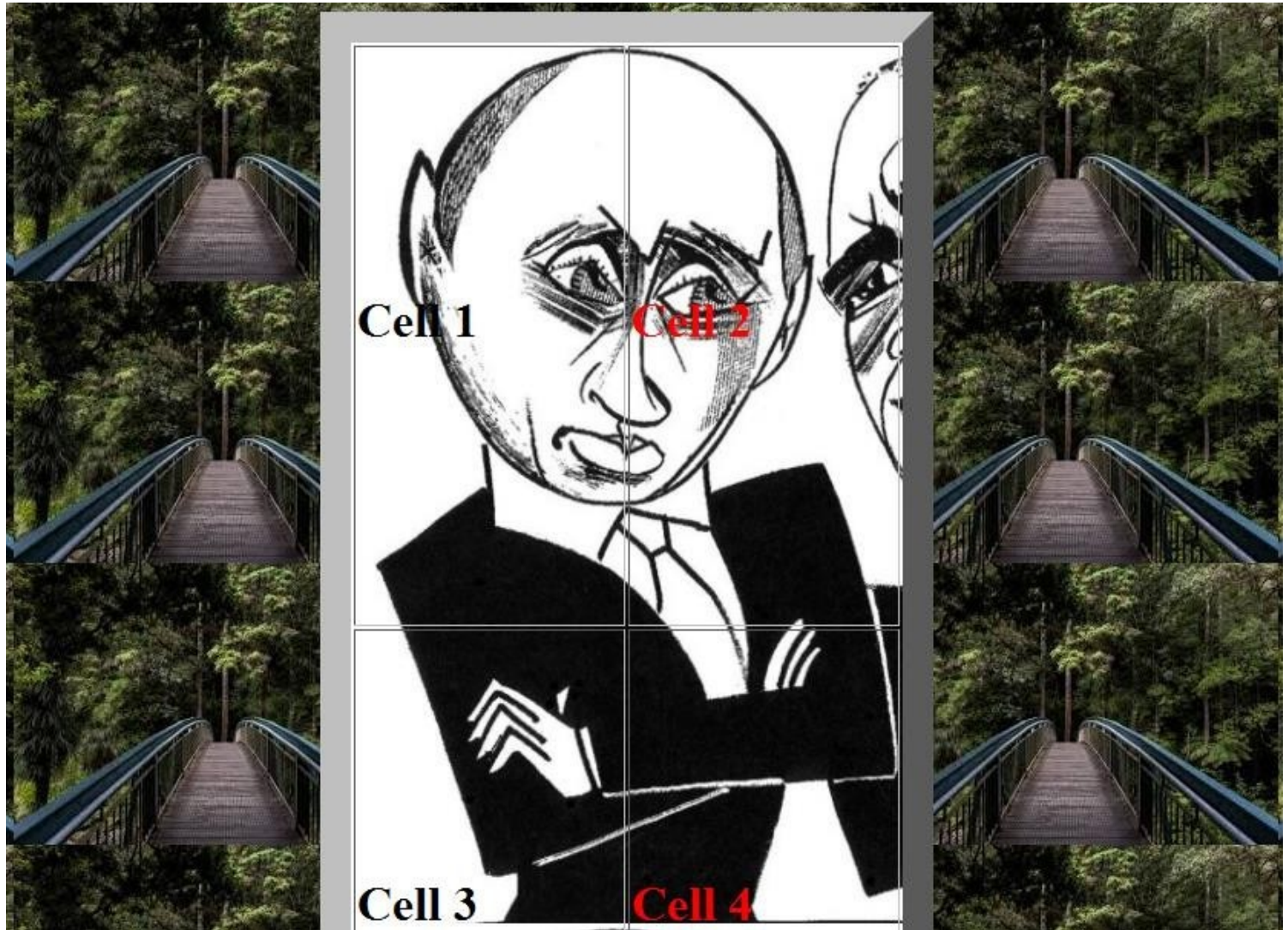
```
</head>
```

```
<body>
```

```
<table border="2"> <tr> <td>Cell 1</td> <td>Cell  
2</td> </tr> <tr>
```

```
<td>Cell 3</td> <td>Cell 4</td> </tr>  
</table> </body> </html>
```

Output: Table with image back ground



CSS background-repeat Property

- The background-repeat property sets if/how a background image will be repeated.
 - By default, a background-image is repeated both vertically and horizontally.
 - background-repeat: repeat|repeat-x|repeat-y|no-repeat;

Example body {background-image:
url(images/ob020.jpg);

background-repeat:no-repeat;
background-position: 30% 20%; }

Background-position

- The **background-position** property sets the starting position of a background image.
 - By default, a background-image is placed at the top-left corner of an element, and repeated both vertically and horizontally.
 - Value can be left top, left center, left bottom, right top, right center, right bottom, center top, center center, center bottom, $x\%$ $y\%$, $xpos$ $ypos$.

Example `body {background-image: url(images/ob020.jpg);
background-position: 30% 20%; }`

background-size Property

- Specify the size of a background image:
 - Background-size: auto|*length*|percent|

contain Example

```
body {  
    background: url(logo.png);  
    background-size: 100% 100%;  
}
```

2. CSS text properties

- ❑ With CSS text properties, you can define how text is displayed. You can specify the color of text, alignment of text, and more.
- ❑ This focuses on:
 - ❑ **Setting text color**
 - ❑ **Aligning text**
 - ❑ **Decorating text**
 - ❑ **Setting letter spacing**

A. Setting text color

- ❑ Text color is set with the **color property**.
- ❑ You can **specify the color of text with a color name, an RGB value, or a hexadecimal value.**
- ❑ Example:



```
<html>
```

```
<head> <title>Background properties</title>
```

```
<style type="text/css">
```

```
h1 {color: green;}
```

```
p {color: #00008b;}
```

```
p.text{color: rgb(47, 49, 49);}
```

```
</style>
```

```
</head>
```

```
<body> <h1>This text is  
green</h1>
```

```
<p>This text is darkblue</p>
```

```
<p class="text">This text is  
darkslategray</p>
```

```
</html>
```

```
</body>
```

Output:

This text is green

This text is darkblue

This text is darkslategray

B. Aligning text

- ❑ The alignment of text in an element is set with the **text-align** property. With this property, you can set text to align to the left, right, or center.

❑ Example

```
<html> <head> <title>Background properties</title>
<style type="text/css">
h1 {text-align: left;}
h2 {text-align: right;}
p {text-align: center;}
</style>
</head> <body>
<h1>This text is aligned to the left</h1> <h2>This text is aligned to
the right</h2> <p>This text is aligned to the center</p> </body>
</html>
```

Output:

This text is aligned to the left

This text is aligned to the right

This text is aligned to the center

C. Decorating text

- ❑ Text decoration is set with the **text-decoration** property.
- ❑ With this property, you can set **text to be underlined, have an over line**, and more.

Example:

```
<html> <head> <title>Background properties</title>
<style type="text/css">
p {text-decoration: underline;}
a {text-decoration: overline;}
h2 {text-decoration: line-through;}
h1 {text-decoration: blink;}
</style>
</head>
<body> <p>This text is
underlined</p>
<a href="http://www.landofcode.com">Landofcode.com
main page</a> <h2>This text has a line through it</h2>
<h1>This text is blinking</h1>
```

```
</body> </html>
```

Output:

This text is underlined

[Landofcode.com main page](#)

~~This text has a line through it~~

This text is blinking

D. Setting letter spacing

- ❑ Letter spacing is set with the **letter-spacing** property.
- ❑ You can increase or decrease the letter spacing between text with this property.
- ❑ The **letter-spacing** property takes the value "**normal**" for normal spacing between text characters or a numeric attribute specified by pixels followed by "**px**" for a fixed space between text characters.

3. CSS font properties

❑ With CSS font properties, you can specify the font a text will be displayed in and the size of the text.

❑ **This focuses on:** Setting text font & Setting text size

A.Setting text font

❑ Text font is set with the **font-family** property.

❑ You can specify a list of fonts with this property and the web browser will use the first font that is found on the user's computer.

Example:

```
<html>
```

```
<head> <title>Background properties</title>
```

```
<style type="text/css">
```

```
h1 {font-family: helvetica, georgia;}
```

```
p {font-family: tahoma, "trebuchet ms"}
```

```
</style>
```

```
</head>
```

```
<body> <h1>Some text</h1> <p>Some more text</p>
```

```
</body> </html>
```

Output:

Some text

Some more text

B. Setting text size

- ❑ Text size is set with the **font-size** property. You can specify a size from a set of prefixed sizes or set your own size for text.
- ❑ **Some prefixed text sizes:** - x-small, small, medium, large, x-large

Example:

```
<html>
  <head> <title>font
</title> <style type="text/css">
h1 {font-size: 40px;}
p {font-size: large;}
</style>
</head>
<body> <h1>Some text</h1>
<p> A sentence in a paragraph.
Another
sentence in a paragraph. </p>
</body>
</html>
```

Output:

Some text
A sentence in a paragraph. Another sentence in a paragraph

4. CSS list properties

- ❑ With CSS list properties, you can specify how lists appear.
- ❑ **This focuses on:** Setting the marker for a list specifying where a list is placed

A. Setting the marker for a list

- ❑ The marker for a list is set with the **list-style-type** attribute.
- ❑ **Possible values for the list-style-type attribute:** disc, square, decimal, lower-roman, upper-alpha

Example:

```
<html>
```

```
  <head>
```

```
    <title>Lists</title>
```

```
  </head> <body>
```

```
    <ul style="list-style-type: square;"> <li>green</li> <li>blue</li>  
    <li>gray</li> <li>orange</li> </ul>
```

```
    <ul style="list-style-type: upper-alpha;"> <li>green</li> <li>blue</li>  
    <li>gray</li> <li>orange</li> </ul>
```

```
    <ol style="list-style-type: disc;"> <li>green</li> <li>blue</li>  
    <li>gray</li> <li>orange</li> </ol>
```

```
    <ol style="list-style-type: circle;"> <li>green</li> <li>blue</li>  
    <li>gray</li> <li>orange</li> </ol>
```

```
  </body>
```

```
</html>
```

Output:

- green
- blue
- gray
- orange

- A. green
- B. blue
- C. gray
- D. orange

- green
- blue
- gray
- orange

- green
- blue
- gray
- orange

An Image as The List Item Marker

- The **list-style-image** property specifies an image as the list item marker:

```
ul {  
    list-style-image:url('sqpurple.gif');  
}
```

5. CSS border properties

- ❑ With CSS border properties, you can specify how borders appear around elements.
- ❑ **This focuses on:** Setting the style of a border, setting the width of a border, setting the color of a border

A. Setting the style of a border

- ❑ The style of a border is set with the **border-style** property.
- ❑ **Possible values for the border-style property:** dashed, dotted, solid, groove, double, outset

Example:

```
<html>
<head> <title>Lists</title>
<style type="text/css">
  h1 {border-style: dashed;}
  h2 {border-style: double;}
  p {border-style: dotted;}
  a {border-style: groove;}
</style>
</head>
<body>
<h1>This text has a dashed border</h1> <h2>This text has a double
border</h2> <p>This text has a dotted border</p> <a href="http://w
ww.yahoo.com">This text has a groove border</a>
</body>
</html>
```

Output:

This text has a dashed border

This text has a double border

This text has a dotted border

This text has a groove border

B. Setting the width of a border

- ☐ The width of a border is set with the **border-width** property.
- ☐ The values you can assign to this property include "thin", "medium", "thick", or a custom value in pixels.

Example:

```
<html> <head> <title>Lists</title>
```

```
<style type="text/css">
```

```
h1 {border-style: dashed; border-width: thin;}
```

```
h2 {border-style: double; border-width: medium;}
```

```
p {border-style: dotted; border-width: thick;}
```

```
a {border-style: groove; border-width: 10px;}
```

```
</style> </head>
```

```
<body> <h1>This text has a dashed border</h1> <h2>This
```

```
text has a double border</h2> <p>This text has a dotted
```

```
border</p> <a href="http://www.yahoo.com">This text has a  
groove border that has a thickness of 10 pixels</a>
```

```
</body>
```

```
</html>
```

Output:

This text has a dashed border

This text has a double border

This text has a dotted border

This text has a groove border that has a thickness of 10 pixels

C. Setting the color of a border

- ☐ The color of a border is set with the **border-color** property.
- ☐ The values you can assign to this property include a color name, an RGB value, or a hexadecimal value.
- ☐ **Example:**

```
<html>
```

```
<head> <title>Lists</title>
```

```
<style type="text/css">
```

```
h1 {border-style: dashed; border-color: green;}
```

```
h2 {border-style: double; border-color: #556b2f;}
```

```
p {border-style: dotted; border-width: rgb(128, 128, 128);}
```

```
a {border-style: groove; border-color: #2f7e87;} </style>
```

```
</head> <body> <h1>This text has a green dashed
```

```
border</h1> <h2>This text has a darkolivegreen double
```

```
border</h2> <p>This text has a gray dotted border</p> <a
```

```
href="http://www.yahoo.com">This text has a teal groove  
border</a>
```

```
</body> </html>
```

Output:

This text has a green dashed border

This text has a darkolivegreen double border

This text has a gray dotted border

This text has a teal groove border

6. CSS margin properties

❑ With CSS margin properties, you can specify how much space there should be around elements.

❑ **This focuses on:**

- ❑ Setting the left margin of an element
- ❑ Setting the right margin of an element
- ❑ Setting the top margin of an element
- ❑ Setting the bottom margin of an element

A. Setting the left margin of an element

- ❑ The left margin of an element is set using the **margin-left** property.

Example:

```
<html>
<head> <title>Setting the left margin of an element</title>
</head>
<body>
<p>This paragraph has no specified margin.</p>
<p style="margin-left: 20px;">This paragraph has a left margin of
20 pixels.</p>
</body>
</html>
```

B. Setting the right margin of an element

- The right margin of an element is set using the **margin-right** property.

Example:

```
<html>
<head> <title>Setting the right margin of an
element</title>
</head>
<body> <p>This paragraph has no specified
margin.</p>
<p style="margin-right: 25px;">This
paragraph has a right margin of 25 pixels.</p>
</body> </html>
```

C. Setting the top margin of an element

❑ The top margin of an element is set using the **margin-top** property. **Example:**

```
<html>
```

```
<head>
```

```
<title>Setting the top margin of an element</title> </head>
```

```
<body> <p>This paragraph has no specified margin.</p>
```

```
<p style="margin-top: 45px;">This paragraph has a top  
margin of 45 pixels</p>
```

```
</body>
```

```
</html>
```

D. Setting the bottom margin of an element

□ The bottom margin of an element is set using the **margin-bottom** property. **Example:**

```
<html>
```

```
<head> <title>Setting the bottom margin of an element</title>
```

```
</head> <body> <p>This paragraph has no specified  
margin.</p>
```

```
<p style="margin-bottom: 35px;">This paragraph has a bottom  
margin of 35 pixels.</p>
```

```
</body> </html>
```

```
<html>
```

```
<head>
```

```
<style type="text/css"> /* this is a  
comment this is another comment */
```

```
p{ color: green; /* the color of the  
text in the paragraph */
```

```
font-family: Courier; margin-top: 0;  
margin-left: 0; /* the left margin of  
the paragraph */ }
```

```
</style>
```

```
</head> <body> <p> The text in  
this paragraph will be green. </p>
```

```
</body>
```

```
</html>
```

Output

The text in this paragraph will be green.

The **display** Property

- The display property is the most important CSS property for controlling layout.
- The display property specifies if/how an element is displayed.
- Value can be **inline**, **block**

Visibility Properties

- The visibility property specifies whether or not an element is visible
- Value can be **visible** and **hidden**

<div>

- The <div> tag defines a division / section
- Also known as “div boxes”
- If you declare a div box to be a height and width, then it will take up that much space on the screen, starting from where you called it
- This is why a lot of people use CSS –you can put anything anywhere you want it on the screen without having to rely on tables.
- For this upcoming example, let’s use an internal style sheet...

```
<html><head>  
<style type="text/css">  
  Div{ Width:1  
  00%  
  Height:80px;  
  Color:white;  
  Background-color:green  
  Font-size:200%  
}</style></head>  
<body>  
  <div> this is some text  
  in side block  
</div></body></html>
```

Div properties

- Float: Float decides where a given element will occur, so, for instance, on the left hand side of the webpage or the right hand side
 - value can be left or right
- Overflow: The overflow property decides what to do if the content inside a given element exceeds the height and width
 - Value can be visible, hidden or scroll
- Position: Value can be relative or absolute
 - Absolute positioning means that an element can be placed anywhere on the web page by simply specifying the number of pixels from the top of the webpage and the number of pixels from the left-hand side of the webpage

Div properties

- **top** specifies how far from the top an element will appear either -in pixels or a percentage
- **left** specifies how far from the left-hand side an element will appear either -in pixels or a percentage.

```
<html><head>
<style type="text/css">
div.right{
background-
color:green;
float:right;
}
div.scrollbars{ backgroun
d-color:yellow; float:left;
overflow:scroll;
width:30%;
height:40%;
}
div.random{ backgrou
nd-color:red;
position:absolute;
top:300px;
left:250px;
}
</style></head>
```

```
<body>
<div class="right">this div is on <br>the
right side</div>
<div class="scrollbars">this div box use
scroll bars</div>
<div class="random">this div box is
300px in
and 150px down</div>
</body></html>
```

7. CSS comments

- ❑ Comments are declared within stylesheet definitions so that code would be easier to understand and navigate. Comments in CSS are ignored by web browsers.
- ❑ Comments begin with `/*` and end with `*/` and can span as many lines as you want.
- ❑ **Example:**

Positioning and Layout Properties:

- These properties allow you to control how elements are displayed and positioned within the browser window or relative to other elements.

Position property : The position property specifies the type of positioning method used for an element(determines how an element is positioned in the document.).

There are five different position values:

- static
 - relative
 - fixed
 - absolute
 - Sticky
- Elements are then positioned using the **top, bottom, left, and right** properties. However, these properties will not work unless the position property is set first. They also work differently depending on the position

Position:

- HTML elements are positioned static by default.
- Static positioned elements are not affected by the top, bottom, left, and right properties.
- An element with position: static; is not positioned in any special way;
it is always positioned according to the normal flow of the page:

Position: relative

- An element with position: relative; is positioned relative to its normal position.
- Setting the top, right, bottom, and left properties of a relatively-positioned element will cause it to be adjusted away from its normal position.

position:

absolute;

- An element with **position: absolute;** is positioned relative to the nearest positioned ancestor
- However; if an absolute positioned element has no positioned ancestors, it uses the document body, and moves along with page scrolling.

position: sticky;

- A hybrid of **relative** and **fixed**.
- The element behaves like relative until it reaches a defined scroll position, then it sticks in place like fixed.

Example

```
<head>
```

```
<style>
```

```
div.static {
```

```
  position:
```

```
  static;
```

```
  border:
```

```
  3px solid
```

```
  #73AD2
```

```
  1;
```

```
}
```

```
div.relative
```

```
{ position:
```

```
relative; left:
```

```
30px;
```

```
width: 400px;
```

```
height: 200px;
```

```
border: 3px
```

```
solid
```

```
#73AD21;
```

```
}
```

```
div.fixed
```

```
{ position:
```

```
fixed; bottom:
```

```
0;
```

```
<body>
```

```
<div class="static">
```

This div element has position: static;

```
</div>
```

```
<div class="fixed">
```

This div element has position: fixed;

```
</div>
```

```
<div class="relative">
```

This div element has position: relative;

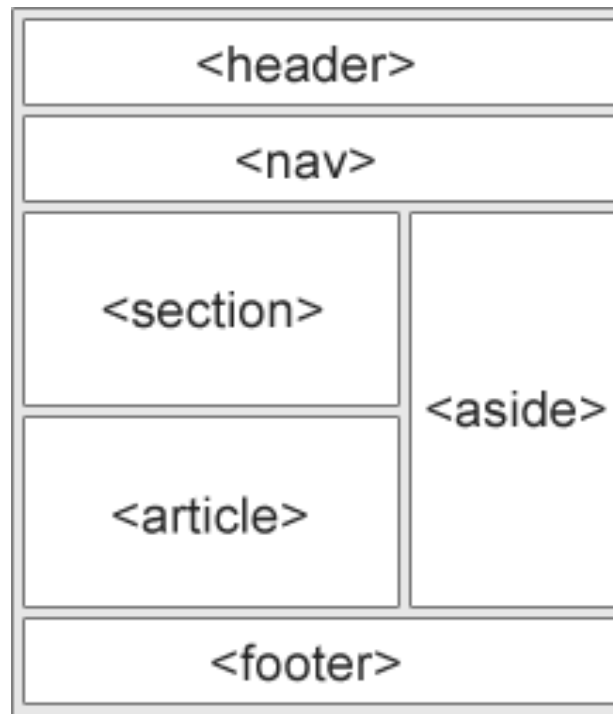
```
</div>
```

```
</body>
```

```
</html>
```

Exercise

- The following image show example of semantic element layout by w3schools position this semantic element using position property



Exercise output

Fixed Header

Main Content

This is the main content area. The header is fixed at the top, and this area scrolls below it.

Notice that the sidebar is absolutely positioned on the right.

Sidebar

This is an absolutely positioned sidebar.

Sticky Footer

Bootstrap

- Bootstrap is a front-end framework used for developing responsive websites.
- Bootstrap is a popular front-end framework used in web development to create responsive and mobile-first websites quickly and efficiently.
- Bootstrap is built using HTML, CSS, and JavaScript.
- It includes HTML, CSS, and JavaScript components that help developers build modern and visually appealing web pages without starting from scratch.

Bootstrap

Bootstrap is primarily used to make web design **faster**, **easier**, and **more responsive**. Here are the key reasons why developers and designers use Bootstrap:

- **Time-saving:** Working on a project that has a number of templates can be time-consuming.
- With Bootstrap Pre-designed CSS components, you get to use pre-made design themes and templates as your starting point. It allows modification of its code to make the framework suit your project's needs. In addition, any changes you make during the development process will also affect all of the webpages altered by Bootstrap.
- **Easy to Use:** Bootstrap is very simple to use. the setup process doesn't take too long and is relatively easy, even for beginners
- **Consistency:** Maintains uniform styling across projects.

Bootstrap

- **Responsive Grid System:** One of today's most significant demands in terms of web development is making sure that websites are **responsive and mobile-friendly**. Bootstrap has that covered with their dedicated mobile-first grid system – it can divide the screen into twelve columns to accommodate various screen sizes, thus making its web design adaptable to all kinds of devices.
- **Customizable:** If you're not happy with Bootstrap's design template, you can add your own twist to its CSS file. You can also combine it with the existing code and make them complement each other's functions.
- **Cross-browser Compatible:** Bootstrap is compatible with the latest versions of all modern browsers and platforms
- **Open Source:** Bootstrap is an open-source framework, meaning that you can use and modify it without purchasing any license

Bootstrap

- Bootstrap comes with a wide range of **pre-designed CSS components** that help developers build attractive and responsive web interfaces quickly.
- These components are styled and functional out of the box, making development faster and more efficient.

Limitation of Bootstrap

Some limitation of Bootstrap

- Bootstrap has **Generic & Similar Designs** it requires additional CSS customization to create a unique design
- **Heavy File Size (Performance Issues)**, Bootstrap comes with a lot of unused CSS & JS, which can slow down page loading
- **Not Ideal for Complex Web Applications**, Bootstrap is great for small projects but can become limiting for large-scale apps
- **Too Many Div Elements ("Div Soup")**, Bootstrap's grid system relies on many nested `<div>` elements, which can make HTML less readable and harder to maintain.
- **SEO & Accessibility Challenges**, Bootstrap sites often use too many HTML elements, which can affect SEO rankings.

Getting Started with Bootstrap

- You can use Bootstrap in two main ways

1. CDN (Content Delivery Network) – Add the following link in your `<head>` section

```
<link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css">
```

2. Download and Install – Download Bootstrap from **getbootstrap.com** or install via npm:

open cmd and type `npm install bootstrap`

Read more about Bootstrap components and check the following simple example

Getting Started with Bootstrap

Simple example using bootstrap

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1">
```

```
  <title>Simple Bootstrap Page</title>
```

```
  <!-- Bootstrap CDN -->
```

```
  <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css">
```

```
</head>
```

```
<body>
```

```
  <!-- Navbar -->
```

```
  <nav class="navbar navbar-expand-lg navbar-dark bg-primary">
```

```
    <div class="container">
```

```
      <a class="navbar-brand" href="#">My Website</a>
```

```
      <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarNav">
```

```
        <span class="navbar-toggler-icon"></span>
```

```
      </button>
```

Getting Started with Bootstrap

```
<div class="collapse navbar-collapse" id="navbarNav">
  <ul class="navbar-nav ms-auto">
    <li class="nav-item"><a class="nav-link" href="#">Home</a></li>
    <li class="nav-item"><a class="nav-link" href="#">About</a></li>
    <li class="nav-item"><a class="nav-link" href="#">Contact</a></li>
  </ul>
</div>
</div>
</nav>
<!-- Hero Section -->
<header class="bg-light text-center py-5">
  <div class="container">
    <h1>Welcome to My Bootstrap Page</h1>
    <p class="lead">This is a simple, responsive webpage using Bootstrap.</p>
    <a href="#" class="btn btn-primary btn-lg">Get Started</a>
  </div>
</header>
<!-- Footer -->
<footer class="bg-dark text-white text-center py-3">
  <p>&copy; 2025 My Website. All rights reserved.</p>
</footer>
</body>
</html>
```